ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

LIFE SCIENCES P1

MARKING GUIDELINE

MARKS: 150

Marking guideline for the Academic Central Grade 10 Life Sciences Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept correct alternatives in the learner's own words. For spelling, accept a phonetically correct spelling unless it has another biological meaning.

SECTION A

1.1 1.1.1 C ✓✓ 1.1.2 B ✓✓ 1.1.3 B ✓✓ 1.1.4 B ✓✓ 1.1.5 B ✓✓ 1.1.6 B ✓✓ 1.1.7 B ✓✓ 1.1.8 B ✓✓ 1.1.9 C ✓✓ 1.1.10 B ✓✓ (20)

1.2 1.2.1 Mitosis ✓ 1.2.2 Chloroplast ✓ 1.2.3 Enzymes ✓ 1.2.4 Tissue ✓ 1.2.5 Phagocytosis ✓ 1.2.6 Amino acids ✓ 1.2.7 Stomata ✓ 1.2.8 Cancer ✓ (8)

1.3 1.3.1 Both A and B ✓✓ 1.3.2 A only ✓✓ 1.3.3 B only ✓✓ 1.3.4 B only ✓✓ 1.3.5 A only ✓✓ 1.3.6 None ✓✓ (12)

1.4.1 A Cell wall ✓ B Cell membrane ✓ C Vacuole ✓ D Nucleus ✓ E Chloroplast ✓ F Mitochondrion ✓ (6)

1.4.2 Controls all the activities of the cell / contains the genetic material (DNA) ✓ (1)

1.4.3 Photosynthesis (traps light energy to make food) ✓ (1)

1.4.4 Any TWO: A, C, E ✓✓ (accept C: animal cells have only small, temporary vacuoles) (2)

TOTAL SECTION A: 50

SECTION B

QUESTION 2

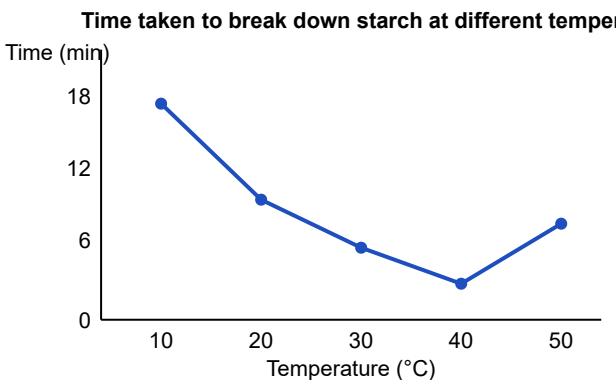
2.1.1 Independent: temperature ✓. Dependent: the time taken for the starch to be broken down ✓ (2)

2.1.2 40 °C ✓ (1)

2.1.3 The high temperature denatured the enzyme ✓ (changed its shape), so it could no longer break down starch ✓. (2)

2.1.4 Any TWO: amount/concentration of starch ✓; amount of amylase ✓; pH; volume of solution; time intervals for testing (2)

2.1.5 (6)



Marks: line graph ✓; correct heading ✓; X-axis label with unit ✓; Y-axis label with unit ✓; correct scale ✓; points plotted correctly ✓ (all 5)

2.1.6 Amylase breaks down starch fastest at an intermediate temperature ✓ and slower at low or very high temperatures ✓ (any testable statement that links temperature to enzyme activity). (2)

2.1.7 Repeat the investigation and calculate averages ✓ (1)

2.2.1 Benedict's solution ✓; heat; colour changes from blue ✓ to orange / brick-red ✓ (3)

2.2.2 Any TWO: growth ✓; repair of tissues ✓; making enzymes and hormones; antibodies (2)

2.2.3 Saturated fats raise cholesterol levels ✓, which can block blood vessels and lead to heart disease / high blood pressure / obesity ✓. (2)

2.2.4 Kwashiorkor ✓ (1)

2.3.1 $\text{X: } \frac{5{,}6 - 5{,}0}{5{,}0} \times 100 \checkmark = +12\% \checkmark \quad \text{Y: } \frac{4{,}1 - 5{,}0}{5{,}0} \times 100 \checkmark = -18\% \checkmark$ (4)

2.3.2 Osmosis ✓ (1)

2.3.3 The salt solution has a lower water potential than the potato cells ✓. Water moved out of the cells ✓ by osmosis through the selectively permeable membranes ✓, so the mass decreased. (3)

2.4 (7)

Plant cell	Animal cell
Has a cell wall	No cell wall ✓✓
Has chloroplasts	No chloroplasts ✓✓
One large central vacuole	Small or no vacuoles ✓✓

Any THREE correct pairs (accept also: regular vs irregular shape; starch vs glycogen). Table with headings ✓

2.5.1 Prophase ✓, metaphase ✓, anaphase ✓, telophase ✓ (in this order) (4)

2.5.2 The centromeres split ✓ and the chromatids (now chromosomes) are pulled to opposite poles of the cell by the spindle fibres ✓. (2)

2.5.3 Any TWO: growth ✓; repair and replacement of damaged or worn-out cells ✓; asexual reproduction (2)

2.5.4 Cancer is uncontrolled cell division (mitosis) ✓ that forms a tumour ✓. Treatment, any ONE: chemotherapy, radiation therapy, surgery ✓ (3)

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QUESTION 3

3.1.1 A Cuticle ✓ B Upper epidermis ✓ C Palisade mesophyll ✓ D Spongy mesophyll ✓ E Vascular bundle (vein) ✓ F Stoma / guard cells ✓ (6)

3.1.2 A: prevents water loss (waterproof layer) ✓. F: allows gas exchange (CO_2 in, O_2 out) / transpiration ✓ (2)

3.1.3 They contain many chloroplasts and are closest to the light ✓, so they can trap as much light as possible for photosynthesis ✓. (2)

3.2.1 (a) Epithelial (squamous epithelium) ✓ (b) Connective tissue (tendons) ✓ (c) Nerve tissue ✓ (d) Epithelial tissue ✓ (4)

3.2.2 Red blood cells ✓, white blood cells ✓, platelets ✓ (3)

3.3.1 Any TWO, each with both sides ✓✓: xylem carries water and minerals / phloem carries food (sugars); xylem is made of dead cells / phloem of living cells; xylem has lignified walls / phloem has sieve tubes with companion cells; xylem transport is upwards only / phloem in both directions. (4)

3.3.2 The long, thin extension increases the surface area for absorption ✓; it has a thin cell wall and membrane for fast osmosis ✓; its vacuole has a low water potential, so water moves in ✓. (3)

3.4.1 12 : 48 ✓ = 1 : 4 ✓ (2)

3.4.2 The lower surface is cooler and shaded from direct sun ✓, so less water is lost through transpiration ✓. (2)

3.4.3 Guard cells ✓ (1)

3.5.1 An unspecialised cell ✓ that can divide and develop into different types of specialised cells ✓. (2)

3.5.2 Any TWO: treating leukaemia (bone marrow transplants) ✓; repairing damaged tissue such as heart muscle or spinal cord ✓; treating diabetes; growing new tissues for burns (2)

3.5.3 Collecting embryonic stem cells destroys the embryo ✓✓, which some people believe is a potential human life (religious / ethical objection). (2)

3.6 **Content (any 12):** The waxy cuticle reduces water loss ✓. The upper epidermis is transparent and lets light through ✓. The leaf is broad and flat, giving a large surface area to absorb light ✓ and thin so gases diffuse a short distance ✓. Palisade cells are column-shaped and tightly packed near the upper surface ✓, with many chloroplasts ✓ to trap light ✓. Spongy mesophyll has large air spaces ✓ that allow gases to move through the leaf ✓. Stomata, mostly on the lower surface, allow CO_2 in and O_2 out ✓, and guard cells open and close them ✓ to control water loss ✓. Veins (vascular bundles) bring water in xylem ✓ for photosynthesis and carry food away in phloem ✓. Moist cell surfaces let gases dissolve and diffuse ✓. (15)
Synthesis (3): Relevance: only information about leaf structure and adaptations ✓; Logical sequence: ideas arranged in a clear order ✓; Comprehensive: covers both photosynthesis and gas exchange ✓.

[50]

TOTAL: 150

END OF MARKING GUIDELINE